

Gabe Schook and the Bellwether Hiring Team
X, The Moonshot Factory
Mountain View, CA

March 16, 2026

Dear Gabe and the Bellwether team,

I am writing to express my strong interest in the **Machine Learning Software Engineering Lead** role on Project Bellwether. Bellwether's focus on turning geospatial and machine learning systems into reliable, customer-facing products for severe weather and climate risk is closely aligned with the work I have led throughout my career.

At the **Texas Advanced Computing Center (TACC)**, where I served as a **Senior Data Scientist and Technical Lead**, I helped develop real-time flood inundation models and disaster-resilience tools in collaboration with state and federal partners. In that setting, model reliability directly influenced planning and emergency-response decisions. That experience shaped how I build machine learning systems: **with strong data foundations, rigorous evaluation, careful deployment, and clear operational accountability**.

More recently, as **Founder and ML Architect at Homecastr**, I built an end-to-end experimentation, training, evaluation, inference, and monitoring stack for large-scale spatial forecasting. I designed the modeling and validation infrastructure from scratch, including probabilistic evaluation workflows intended to ensure forecast reliability before production release. I have led work across data systems, model development, deployment, API-backed delivery, and production observability, with **hands-on experience operating ML infrastructure on Google Cloud and related tooling**.

What particularly attracts me to Bellwether is the opportunity to help guide production machine learning for real-world geospatial products in a small, highly collaborative team. I am comfortable moving between architecture, modeling, infrastructure, debugging, and stakeholder communication, and I value the combination of technical depth, execution speed, and product ownership that this role requires.

I would be excited to bring my background in **production geospatial ML, high-stakes weather and hazard modeling, and end-to-end ML systems leadership** to Bellwether. Thank you for your time and consideration. I would welcome the opportunity to discuss how I could contribute to the team.

Sincerely,

Daniel Hardesty Lewis

Attached: resume

Daniel Hardesty Lewis
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Daniel Hardesty Lewis

Machine Learning SWE Lead • GCP, MLOps, geospatial ML, observability, distributed systems

Specializing in automated validation frameworks and scalable distributed infrastructure. Delivered consumer pricing and statewide resiliency platforms across geospatial & time-series workloads.

Core Expertise

- 8 years **ML Engineering**: Build and operate AI/ML training and inference pipelines, using rigorous validation protocols to ensure model generalization and reliability
- 8 years **Backend & Data**: Build schemas and ETL; ship API-backed services using Python, SQL, and Postgres/PostGIS
- 8 years **Distributed Systems**: Scale batch workloads across distributed systems, profile bottlenecks, and optimize latency & throughput with explicit cost-performance tradeoffs
- 10 years Production Geospatial ML**: Earth observation, spatiotemporal risk modeling, geospatial ETL, and decision-support systems for hazard and planning workflows.
- 8 years **Technical Leadership**: Drive cross-functional execution via specifications and governance; mentor engineers through design reviews, code reviews, and instruction

Technologies

Modeling	PyTorch; TensorFlow; scikit-learn; Transformers; LoRA/QLoRA/PEFT; Ray; Spark; Dask
LLM apps	embeddings; RAG; LangChain; LangGraph; DSPy; FAISS; Elasticsearch
Data stack	Python; SQL; Pandas; Polars; PyArrow; Parquet; Postgres; PostGIS; Redis; Supabase
Infrastructure	Linux ; Docker; Kubernetes; Modal; Google Cloud Platform (GCP / GCE / GCS) ; Airflow; CI/CD
Additional	CUDA; C++; TypeScript; React; Bash; Git

Experience

Professional

- Nov. 2024 – Present **Co-founder**, *Homecastr*, New York, NY
 - Consumer real-estate pricing and forecasting platform with interactive uncertainty fancharts and temporal leakage-safe expanding-window evaluation.
 - Architected a nationwide neighborhood-level diffusion forecasting model, **delivering highly-calibrated probabilistic forecasts** and maintaining a **stable 25% MdAE over a 4-year horizon**.
 - Engineered architecture incorporating spatial inducing tokens, per-horizon loss normalization, and DDIM sampling to balance predictive precision with calibrated price trajectories.
 - Developed a probabilistic evaluation suite optimizing interval coverage and sharpness to rigorously validate forecast reliability before production delivery.
 - Owned the end-to-end production ML stack from ingestion, training, evaluation, and inference through API-backed real-time delivery and observability.
 - Refined product strategy through direct feedback from venture partners (a16z, MetaProp, Forum Ventures), enterprise leaders (Mapbox, HR&A), and end-users.
- Nov. 2023 – Present **Founder**, *Summit Geospatial*, New York, NY
 - High-resolution elevation data and web delivery for engineering and hazard workflows.
 - Engineered a **statewide seamless Texas DEM mosaic**, resampling 0.5 m LiDAR sources to 1.2 m via nearest-neighbor across **70+ state, federal, and local elevation datasets**.
 - Developed the web distribution platform end-to-end including **data pipeline, tiling, hosting, delivery UX** to support engineering and planning use cases.
- Nov. 2024 – Present **Co-founder**, *PoliBOM*, Seattle, WA
 - AI trade compliance platform for manufacturers, enabling automated BOM retrieval and policy scenario simulation**.
 - Architected an agentic RAG workflow (Airflow, Elasticsearch, FastAPI) and conversational AI interface for instant BOM retrieval, tariff simulation, and compliance alerting.
 - Executing customer discovery with ScoutyAI and Columbia economists, and secured early market validation by presenting the prototype to KPMG.

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- Aug. 2021 – **Senior Data Scientist & Technical Lead**, *Texas Advanced Computing Center (TACC)*, Austin, TX
- Aug. 2023
- Spun out Summit Geospatial alongside UT's VPs of IP to commercialize state-level terrain models from the \$40M [Texas Disaster Information System \(TDIS\)](#), delivering precision decision-support for civil engineering.
 - Developed inter-agency RFP and MOU frameworks with state and federal partners, translating technical requirements into scope and milestones facilitating [Real-Time Inundation Mapping](#) and TDIS.
 - Scaled climate and flood models on supercomputers, executing 800,000-node distributed jobs while managing \$1M+ compute budgets and federal partnerships with NOAA, NSF, USACE, GLO.
 - Developed efficient methods to produce high-resolution 1m flood maps from National Water Model outputs for Texas Emergency Management's Real-Time Inundation Mapping.
 - Partnered with the US Army Corps of Engineers to statistically model compound flood hazards in coastal Texas, integrating high-resolution topographic data with hydrological models.
 - Contributed fundamental research supporting Paola Passalacqua's [2022 EGU Bagnold Medal](#), one of geomorphology's highest honors.
 - Served as Technical Reviewer for [Frontera Doctoral Fellows](#) (2023) and [Large-Scale Computing Pathways](#) (2022) applications, evaluating \$10M+ in computing resource allocations on one of the world's largest supercomputers.
- Feb. 2018 – **Data Scientist & Research Engineer**, *Texas Advanced Computing Center (TACC)*, Austin, TX
- Jul. 2021
- Competed in [DARPA World Modelers](#) to improve disaster resiliency and food security modeling in East Africa.
 - Mentored Petrobras engineers in deep learning and HPC to enable institutional technology transfer.
 - Received research recognition from AAAI President Yolanda Gil during her [farewell address](#) for contributions to automated scientific modeling.

Research

- May 2025 – **Research assistant to Dir. Electrical Engineering Zoran Kostic**, *Columbia University*, New York, NY
- Aug. 2025
- Math benchmarking and finetuning of Deepseek's GRPO algorithm
- Jan. 2025 – **Research assistant to Dir. Financial Engineering Ali Hirsra (Industry-sponsored research)**, *Columbia University*, New York, NY
- Oct. 2025
- Developed a multi-asset CVAE latent factor model with Skew-T Mixture priors; achieved 85% R^2 vs. commercial SaaS (75%) and Fama-French (58%) on backtested holdout.
 - Derived a tractable, axiom-compliant per-feature SHAP algorithm using Hessian-based recursive clustering to provide stable temporal regime-dependent attributions for deep learning models.
- July 2019 – **Summer research intern to Pres. AAAI Yolanda Gil**, *University of Southern California*, Los Angeles, CA
- Aug. 2019
- Integrated hydrological models into the [MINT Platform](#) during a year-long competition with MITRE to provide DARPA the "Airflow for science"

Teaching

- Jan. 2018 – **Co-instructor**, *The University of Texas at Austin*, Austin, TX
- Dec. 2020
- Helped design and teach graduate-level courses: *Machine Learning for the Geosciences* for Petrobras engineers, *Intelligent Systems for the Geosciences*, and *Scientific Computation*.

Select Publications

- 2021 Gil, Y., et al. (incl. **D. Hardesty Lewis**), "Artificial Intelligence for Modeling Complex Systems: Taming the Complexity of Expert Models to Improve Decision Making". *ACM TIS*, 2021. DOI: [10.1145/3453172](#)
- 2019 Garijo, D., et al. (incl. **D. Hardesty Lewis**), "An intelligent interface for integrating climate, hydrology, agriculture, and socioeconomic models". *ACM IUI '19*, 2019. DOI: [10.1145/3308557.3308711](#)

Education

- expected 2026 **M.S., Urban Planning**, *Columbia University*, New York City, NY
- 2017 **B.S., Pure Mathematics**, *University of Texas*, Austin, TX

Key Projects

- oppsAlert Developed the first property- and owner-level ML model of NIMBYism, using 15+ years of 15,000+ points of individual homeowner protest data provided by the City of Austin.

Activities

- Sailing Volunteer Atlantic SAR of the [Cheeki Rafiki](#) Bikepacking US West & East Coasts

Languages

- Spanish Native Proficiency French Professional Working Proficiency

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